Exact-atom adjudication

the true $k = 10$ window is EXACT_DEAD;
float64-log is not the cause

Stefan Hamann

August 30, 2026

Abstract

r446 proved the selected $k = 10$ ABD break is REAL on float64 *atoms*: mpmath verified the recurrence, not the input. The flip pivot $\eta \approx -8 \cdot 10^{-14}$ is a 10^{-6}-relative cancellation after 3788 steps, so atom error of rel $\sim 10^{-16}$ was a live hypothesis. This note rebuilds the $k = 10$ window (k_z197, $N = 4071$) in mpmath (dps 50 and 70) from exact integer prime powers ($u = \mathrm{mp.log}(n)$, tent/arch/fold/DFT all mp). Positions agree with float64 to $4.4 \cdot 10^{-16}$ (one ulp); the chain still flips at $n = 3788$, bit-identical at dps 50 and 70. The 2^k family is ABD-dead past $k = 9$ on its true atoms. No RH claim.

Status. No new SATZ. Propositions 1–2 are a named infra census. Ausgang `EXACT_DEAD / ATOM_ULP_ONLY / FLIP_STABLE_3788 / BAND_ENDS / SLICE_FLOOR_STANDS / THIS_FAMILY_FREQUENTLY_FAI` No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Exact builder

Integer mesh (M, L, N_w) is locked to `V.window_shape` (the r397 family). Values are mp from the start: $u_i = \log n_i$ on exact prime powers $n \le 400000$ (the table cap), tent lags, archimedean GL-48 (nodes refined from float seeds to working dps), circulant DFT by Bluestein, cosine fold. Border: sealed smooth comb $\Delta u = 0.01$ with the same arch+DFT and the PIK fold. No float64 cast into the chain.

Proposition 1 (Atoms are ulp-close)**.** *At $k = 5, 9, 10$ and k_z137, exact-atom positions differ from the float64 builder by* $\mathrm{rel}_x = 4.4 \cdot 10^{-16}$ *(one ulp of* cos*). $k = 10$ weights:* $\mathrm{rel}_w = 6.3 \cdot 10^{-15}$*. The named suspects (float64-log, float64-fold) do not produce a 10^{-6} error.*

Build wall at $k = 10$: window 26 s + border 25 s.

2 The decision

Proposition 2 (EXACT_DEAD)**.** *The $k = 10$ chain on exact atoms at dps 50 (full, 530 s) flips first at $n = 3788$, $\eta = -8.853149582220489 \cdot 10^{-14}$, $\mathrm{rel} = 1.2559886185044 23 \cdot 10^{-6}$, $n_{\mathrm{flip}} = 115$. The same first pivot at dps 70 (to $n = 3793$) is bit-identical. Hence the window is EXACT_DEAD: the sign is a property of the exact-source construction, not of float64 atoms or of recurrence rounding.*

$n/N = 3788/4071 \approx 0.931$: the signed $\tilde{\mu}$-chain dies at the same relative depth as in r445/r446. Divergence versus the float-atom chain first crosses rel 10^{-8} at $n = 218$ (absolute $\sim 10^{-15}$, same sign); there is no sign-split — both constructions flip at $n = 3788$.

$k = 9$ exact-atom prefix 80: 0 flips, η matches float (the living side of the band is consistent). k_z137 exact atoms match to one ulp; the window stays the first dead neighbour of the r446 scan.

3 Consequence

No δ_{10} (the window is not ABD-living). The r445 slice $k = 5..9$ with live $k = 8$ and its floor fit are unchanged ($\delta_\infty = +0.02741$).

On *this* Selected family ($a_k = 2^k$, the r397 mesh), infinitely many ABD-living windows do not exist: the living band ends at $k = 9$ / k_z136, and exact atoms do not revive $k = 10$ or k_z137. The frequently-quantifier $\exists^{\mathrm{f}} k : R^\dagger(W_k) \succeq \frac{1}{2} I$ is therefore not fed by this family. Whether a *different* cofinal family (other a_k, other legal mesh) stays ABD-living is a reviewer fork, not a repair of $a_k = 2^k$.

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis. The identities are a finite recurrence on a named atom construction. The frequently-quantifier gap stays open for other families.

References

[1] Standalone measures; von Mangoldt table.

[2] Selected sequence; $a_k \to \infty$, $\Delta_k \to 0$.

[3] Deep builder; $k = 10/11/12$ not ABD-living.

[4] Float64-atom REAL; last living k_z136.